

Principles of Service Orientation

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Fundamental Service Oriented Architecture (SOA)

- A distinct approach for **separating concerns**
- Logic required to solve a large problem by **decomposing** it into a collection of smaller, related pieces
- Each of these pieces addresses a **concern** or a specific part of a problem

Service Oriented Architecture (SOA)

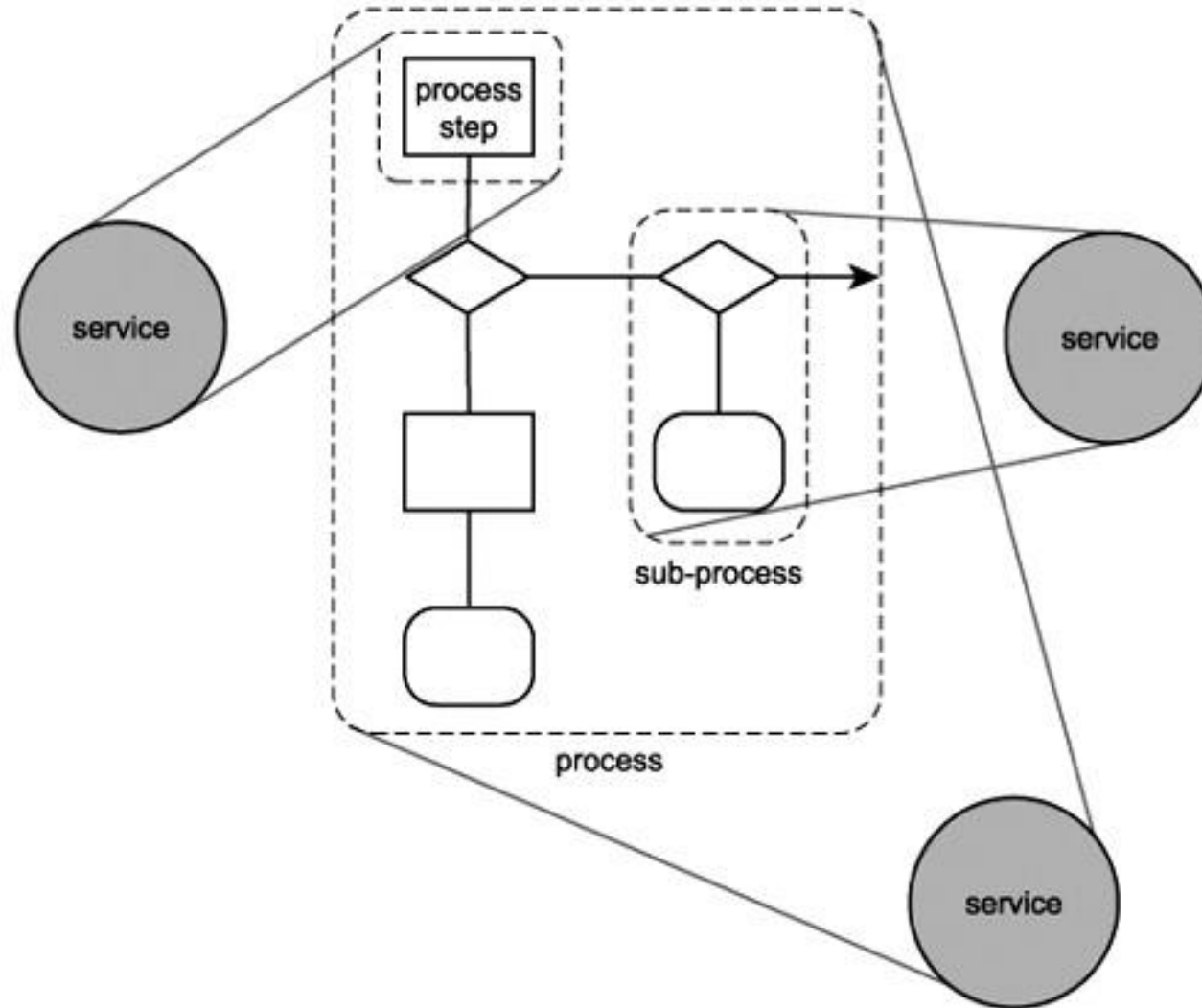
- A model in which automation logic is decomposed into smaller, distinct units of logic
- Individually, these units can be distributed

How Services Encapsulate Logic

- Services **encapsulate logic** within a **distinct context**
 - This context can be specific to a business task, a business entity, or some other logical grouping
- The **concern** addressed by a service can be **small** or **large**
 - The **size and scope** of the logic represented by the service **can vary**
 - **Service logic** can **encompass** the logic provided by other services
- A service can **encapsulate**
 - a **task** performed by an individual step or
 - a **sub-process** comprised of a set of steps or even
 - encapsulate the **entire process logic**

include = encompass

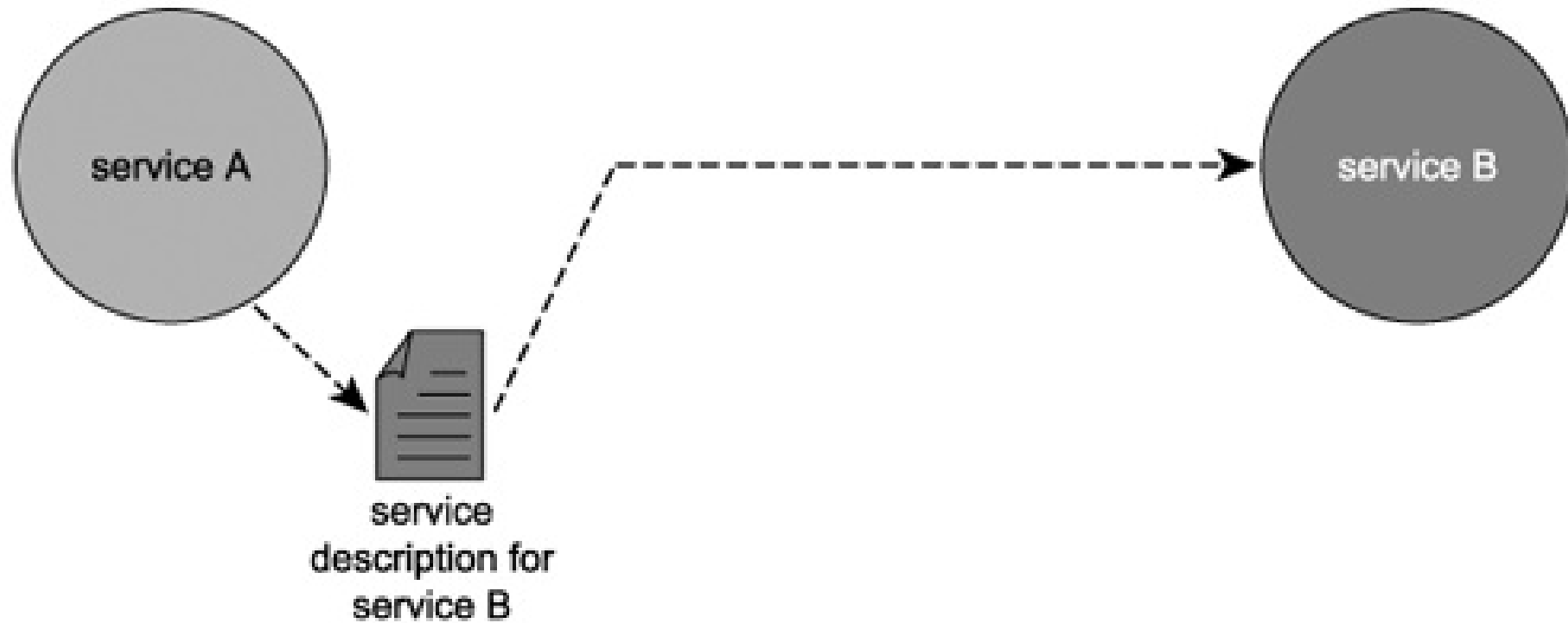
How Services Encapsulate Logic



How Services Relate

- Services can be used by other services or other programs
- Services must be aware of each other
- This awareness is achieved through the use of service descriptions
- A service description in its most basic format establishes the name of the service and the data expected and returned by the service
- The manner in which services use service descriptions results in a relationship classified as loosely coupled

How Services Relate

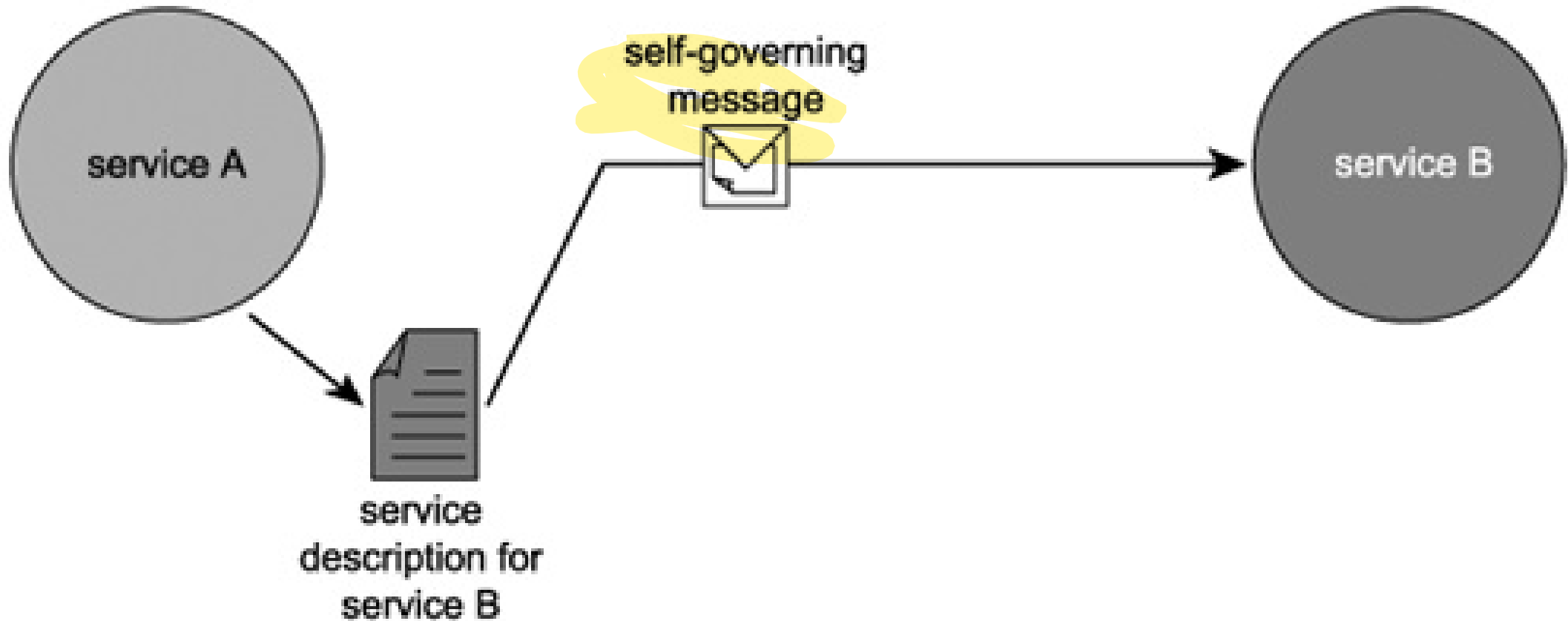


How Services Communicate

- After a service sends a message on its way, it loses control of what happens to the message thereafter
- Messages exist as "independent units of communication."
- Messages, like services, should be autonomous
- Messages can be outfitted with enough intelligence to self-govern their parts of the processing logic

Self-governing intelligence means that the packages themselves have some built-in smarts. They know what to do and where to go without needing constant oversight.

How Services Communicate



How Services are Designed

- Service-orientation principles address the design issues
 - How should services be designed?
 - How should service descriptions be designed?
 - How should messages be designed?
 - How should the relationship between services be defined?

How Services are Built

- Web services technology platform has been found to be suitable and successful in manifesting SOA

Primitive SOA

- What we have described so far about the SOA is called the primitive SOA
- It represents a baseline technology architecture which is supported by current major vendor platforms

Contemporary SOA

- What we have described SOA so far, is called the primitive SOA
- It represents a baseline technology architecture which is supported by current major vendor platforms
- Contemporary SOA builds upon the primitive SOA model by leveraging industry and technology advancements to further its original ideal
- Contemporary SOAs have a set of common characteristics

Characteristics of Contemporary SOA

- 1) It is at the **core** of the **service-oriented computing platform**
- 2) It **increases quality of service**
- 3) It is **fundamentally autonomous**
- 4) It is **based on open standards**
- 5) It **supports vendor diversity**
- 6) It **fosters intrinsic interoperability**
- 7) It **promotes discovery**
- 8) It **promotes federation**
- 9) It **promotes architectural composability**
- 10) It **fosters inherent reusability**

Characteristics of Contemporary SOA

- 11) It emphasizes extensibility
- 12) It supports a service-oriented business modeling paradigm
- 13) It implements layers of abstraction
- 14) It promotes loose coupling throughout the enterprise
- 15) It promotes organizational agility
- 16) It is a building block
- 17) It is an evolution
- 18) It is still maturing
- 19) It is an achievable ideal

1st Generation Web Services Framework Standards

Transports	Messaging	Description	Interoperability
HTTP/HTTPS	SOAP Specification	UDDI	WS-I Basic Profile
	SOAP Attachment	WSDL	WS-I Simple SOAP Binding Profile
			WS-I Attachments Profile

2nd Generation Web Services Framework Standards

XML	Messaging	Reliable Messaging	Metadata Exchange	Security
XML	SOAP	WS-ReliableMessaging	JSON-WSP	WS-Security
XML Namespace	SOAP over UDP	WS-Reliability	WS-Policy	XML Signature
XML Schema	WS-Notification		WS-Discovery	XML Encryption
XPath	WS-Addressing		WS-MetadataExchange	XKMS
XQuery	WS-Transfer		UDDI	WS-SecureConversion
XInclude	WS-Eventing		WSDL	WS-SecurityPolicy
XML Pointer	WS-Enumeration		WSRF	WS-Trust
XML Information set	WS-Connection			WS-Federation
	JMS			SAML

2nd Generation Web Services Framework Standards

Resource	Business Process	Transaction	Management	Privacy
WS Resource Framework	WS-BPEL	WS-BusinessActivity	WS-Management	P3P
WS-Transfer	WS-CDL	WS-AtomicTransaction	WS-Management Catalog	
WS-Fragment	WSCI	WS-Coordination	WS-ResourceTransfer	
	WS-Choreography	WS-Transaction	WSDM	
	WSCL	WS-CAF		
		WS-Context		

SOA Definition

“SOA is a form of technology architecture that adheres to the principles of service-orientation. When realized through the Web services technology platform, SOA establishes the potential to support and promote these principles throughout the business process and automation domains of an enterprise”

Principles of SOA

- Services are

1. reusable

2. share a formal contract

Services in SOA define a formal contract that specifies their interface, including input parameters, output formats, and communication protocols. This contract is shared with service consumers, ensuring clear expectations and interoperability

3. loosely coupled

4. abstract underlying logic

5. composable

Services can be combined and orchestrated to fulfill specific business requirements,

6. autonomous

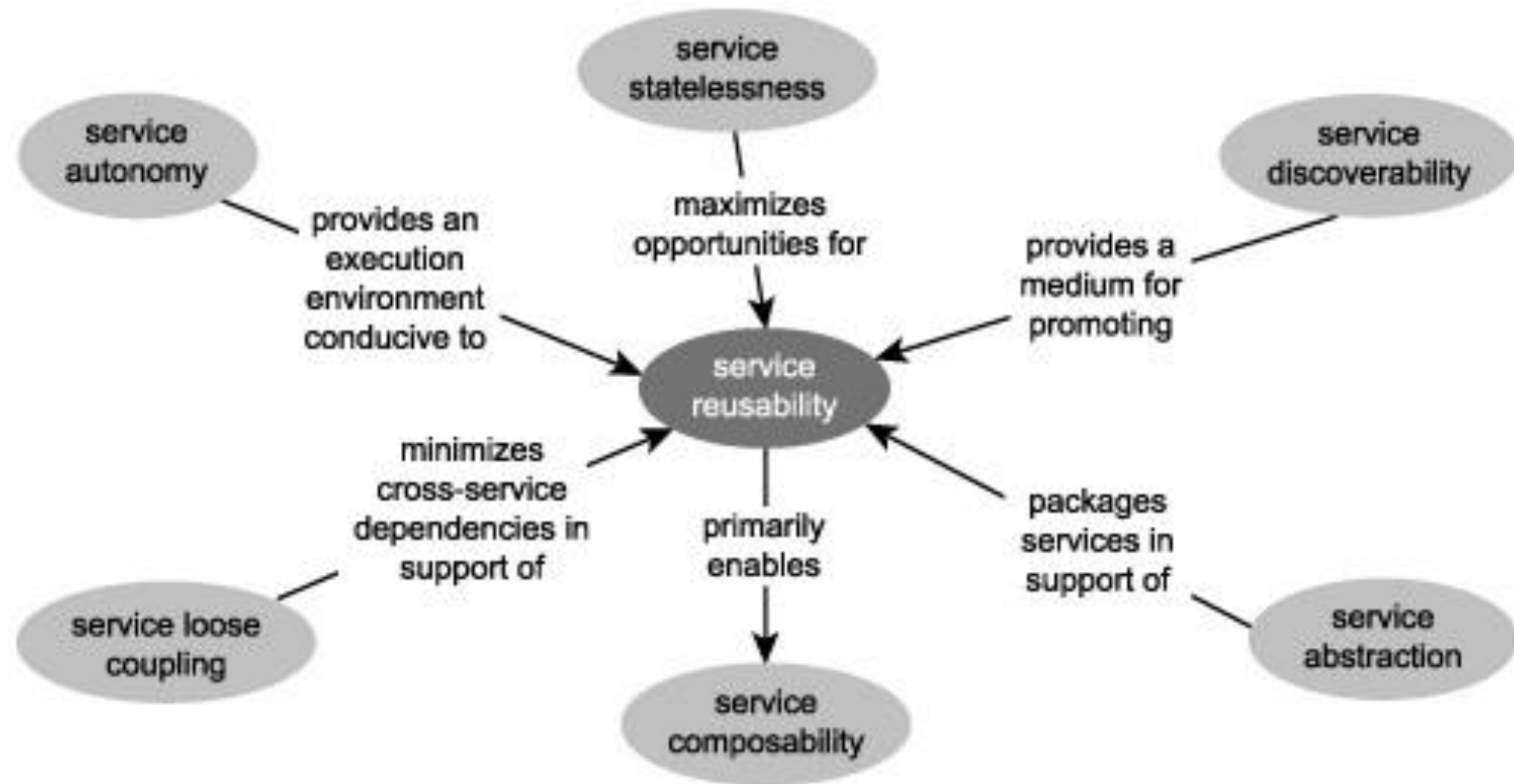
“autonomous” refers to the design principle that services should be independent and self-contained. This means that each service is developed to perform a specific task and operates independently of other services.

7. stateless

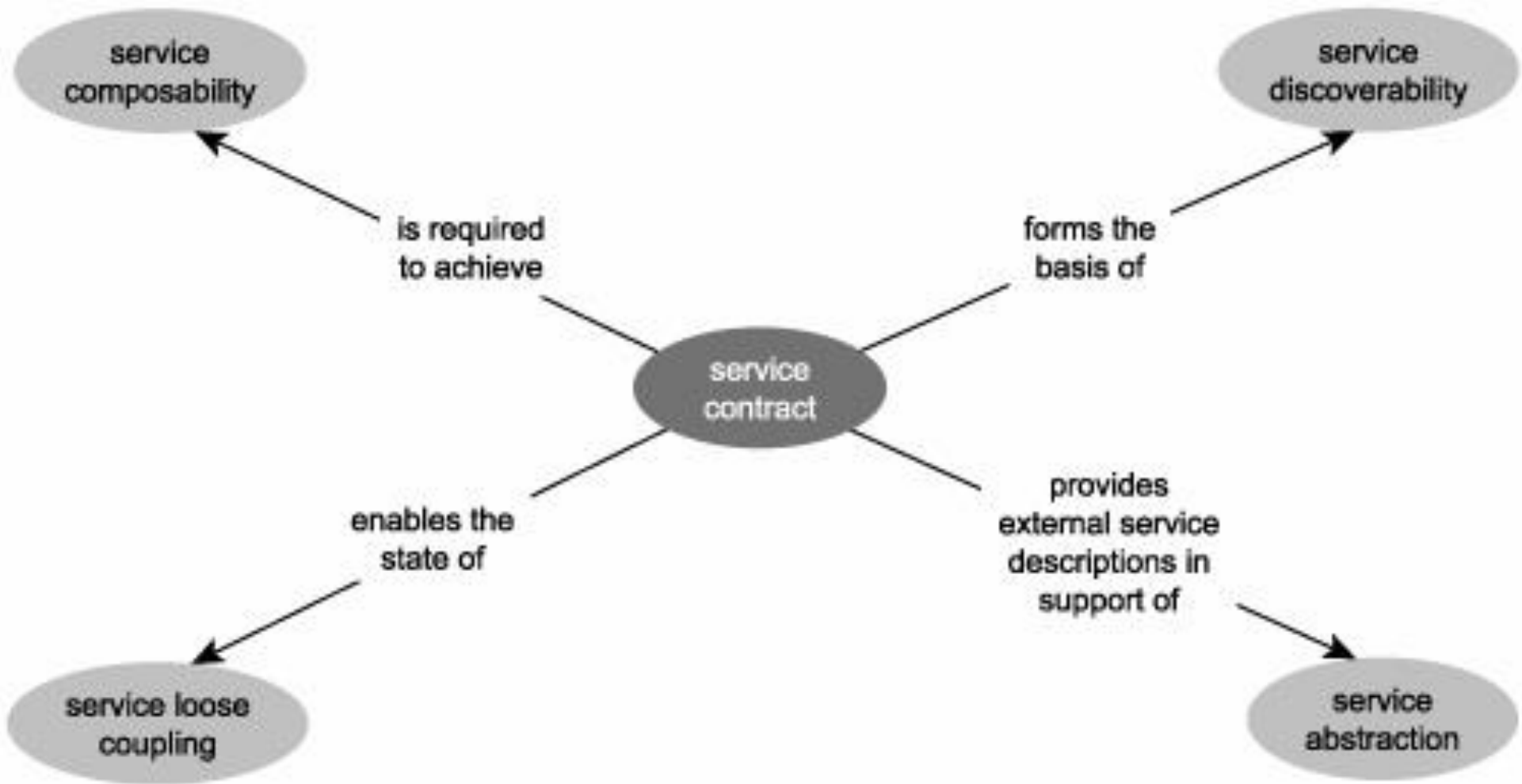
Stateless services in SOA do not maintain session state between interactions. Each request is treated independently, without relying on context from previous interactions

8. discoverable

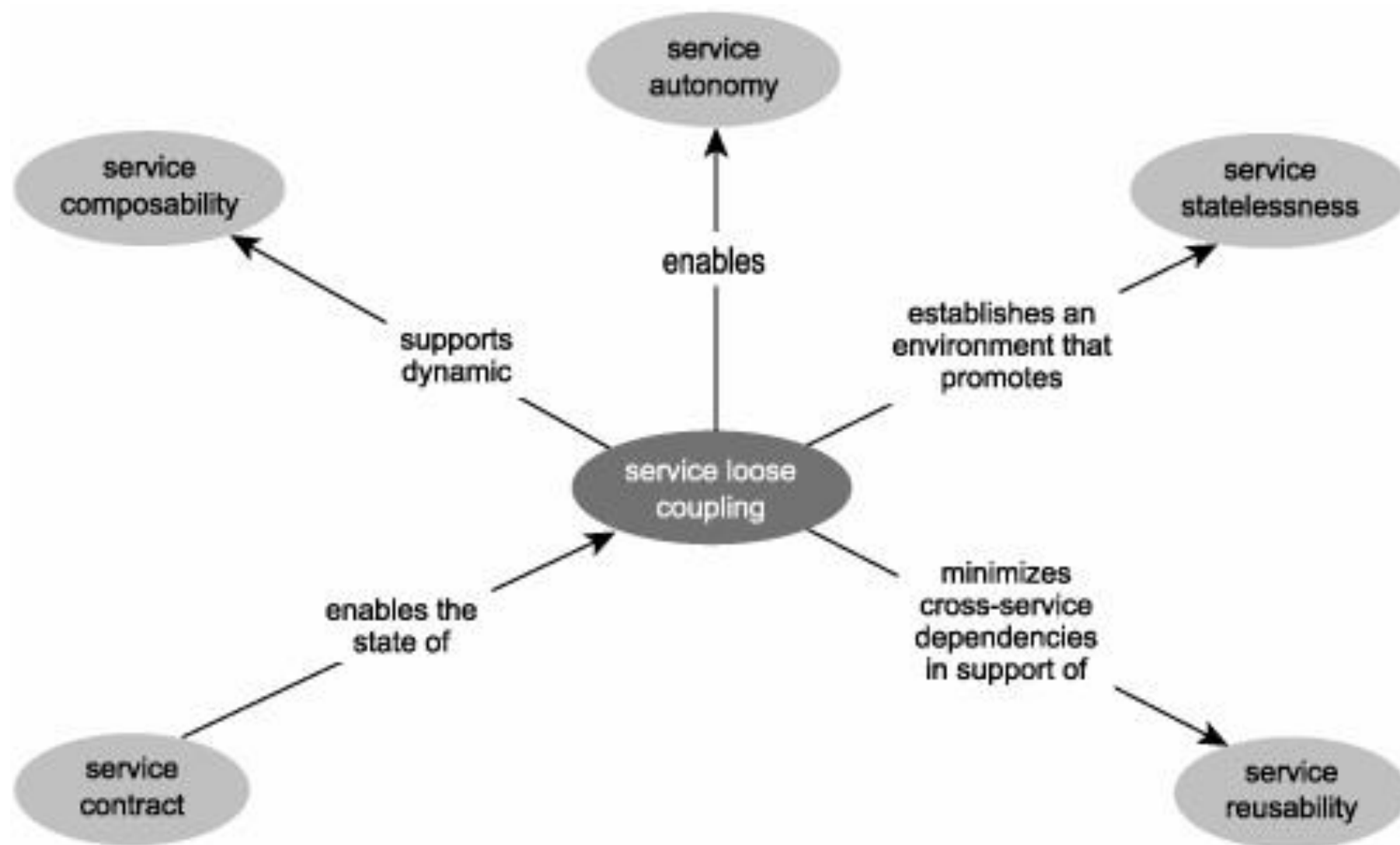
Inter-relation between Service Orientation Principles



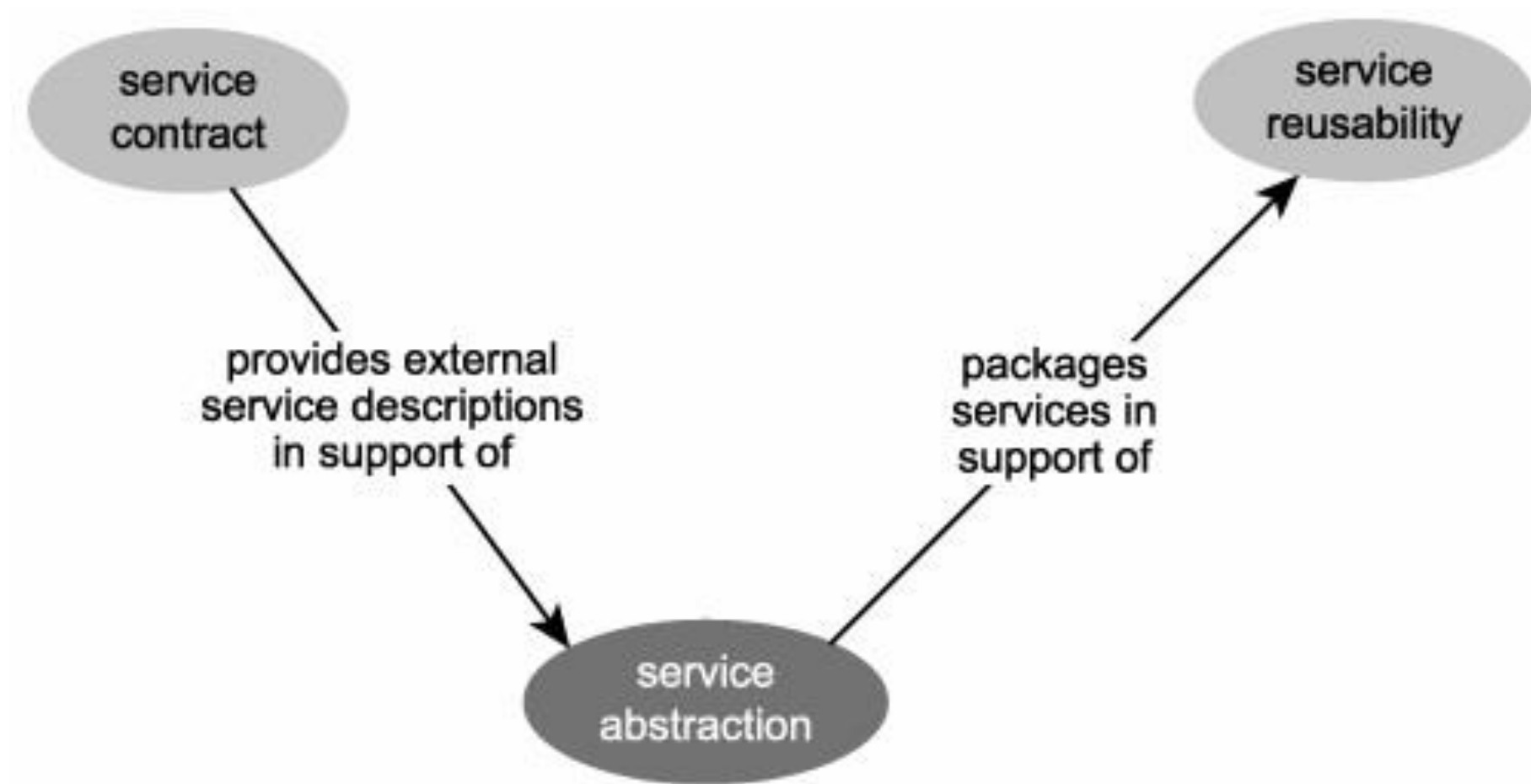
Inter-relation between Service Orientation Principles



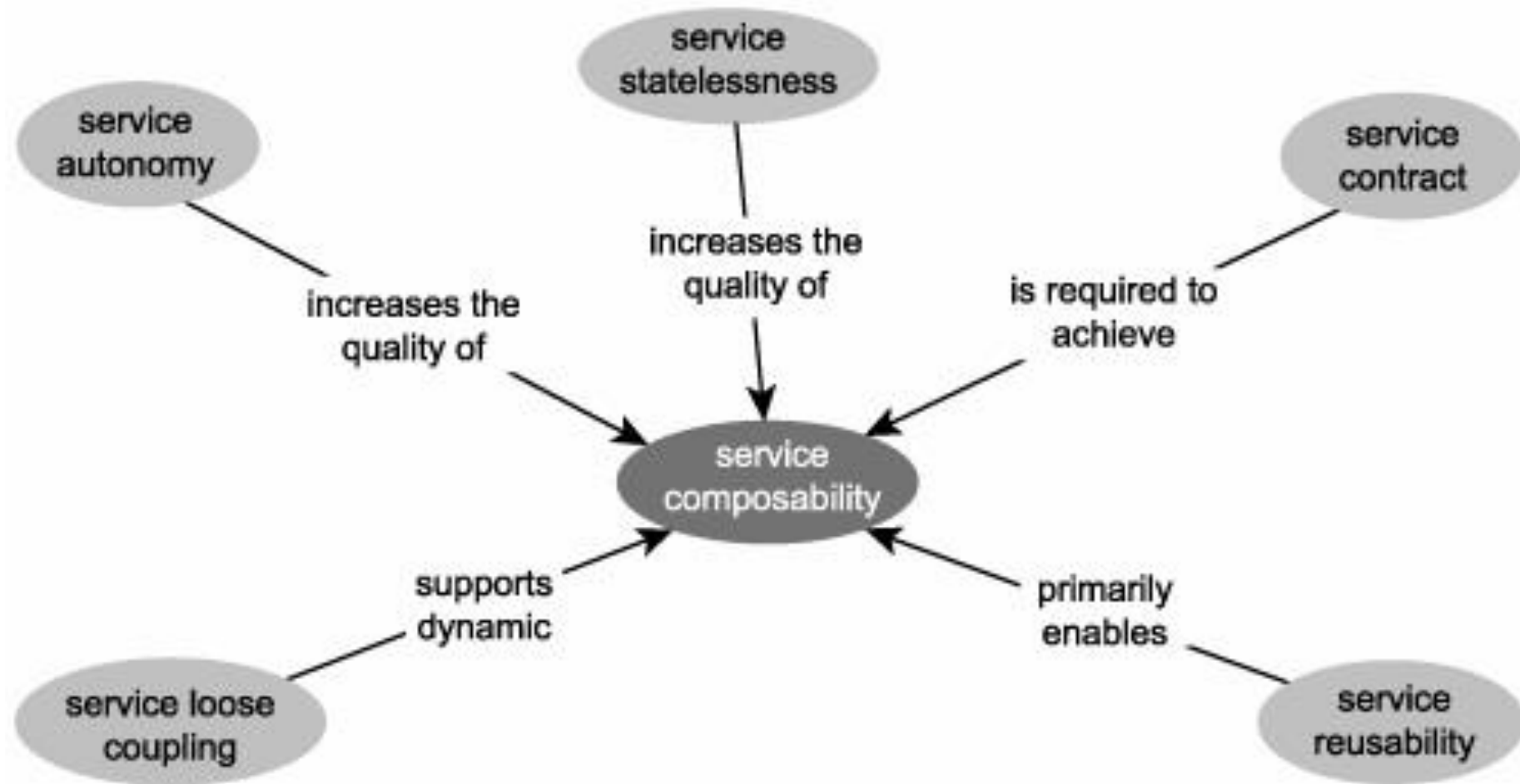
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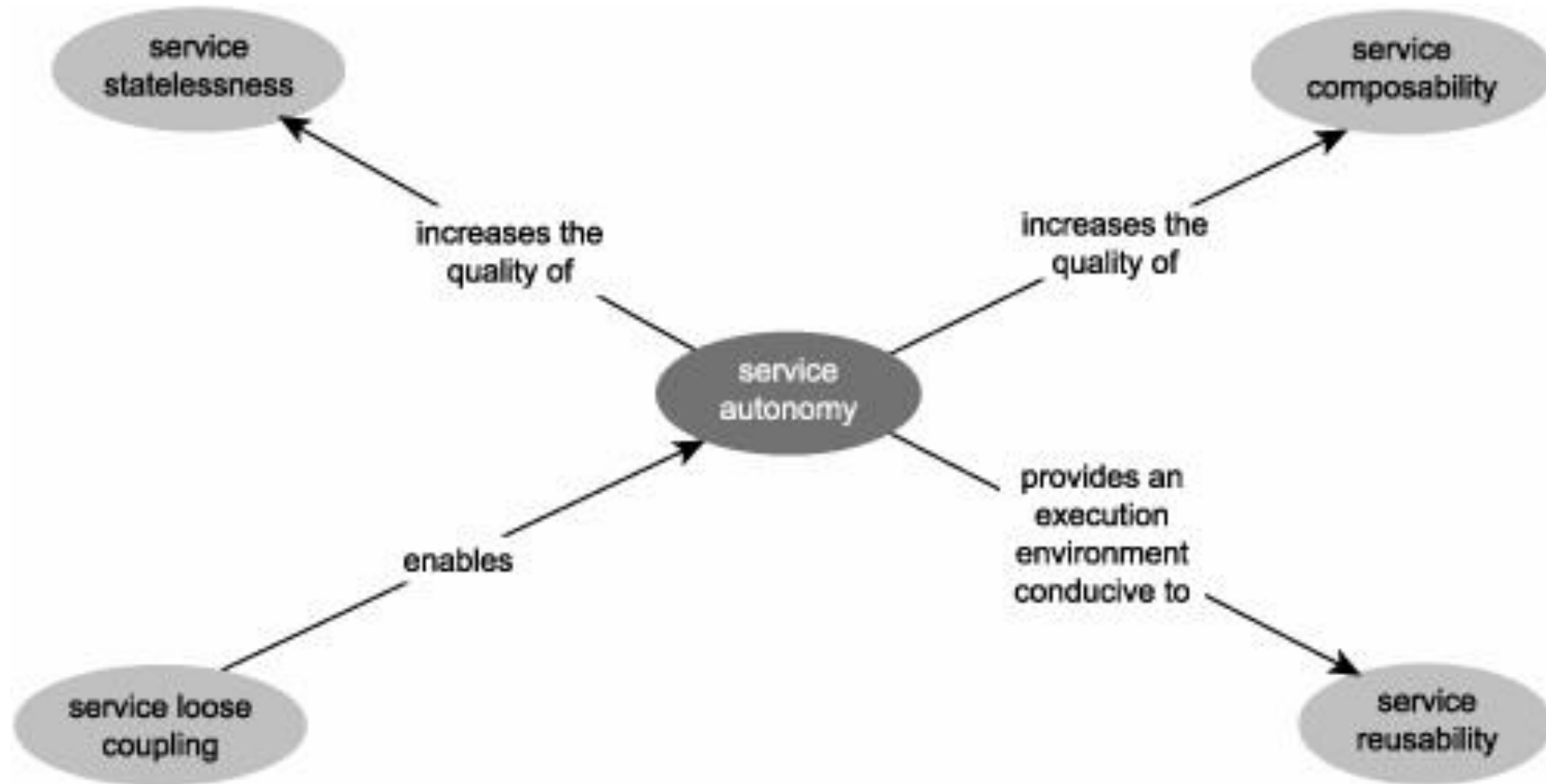
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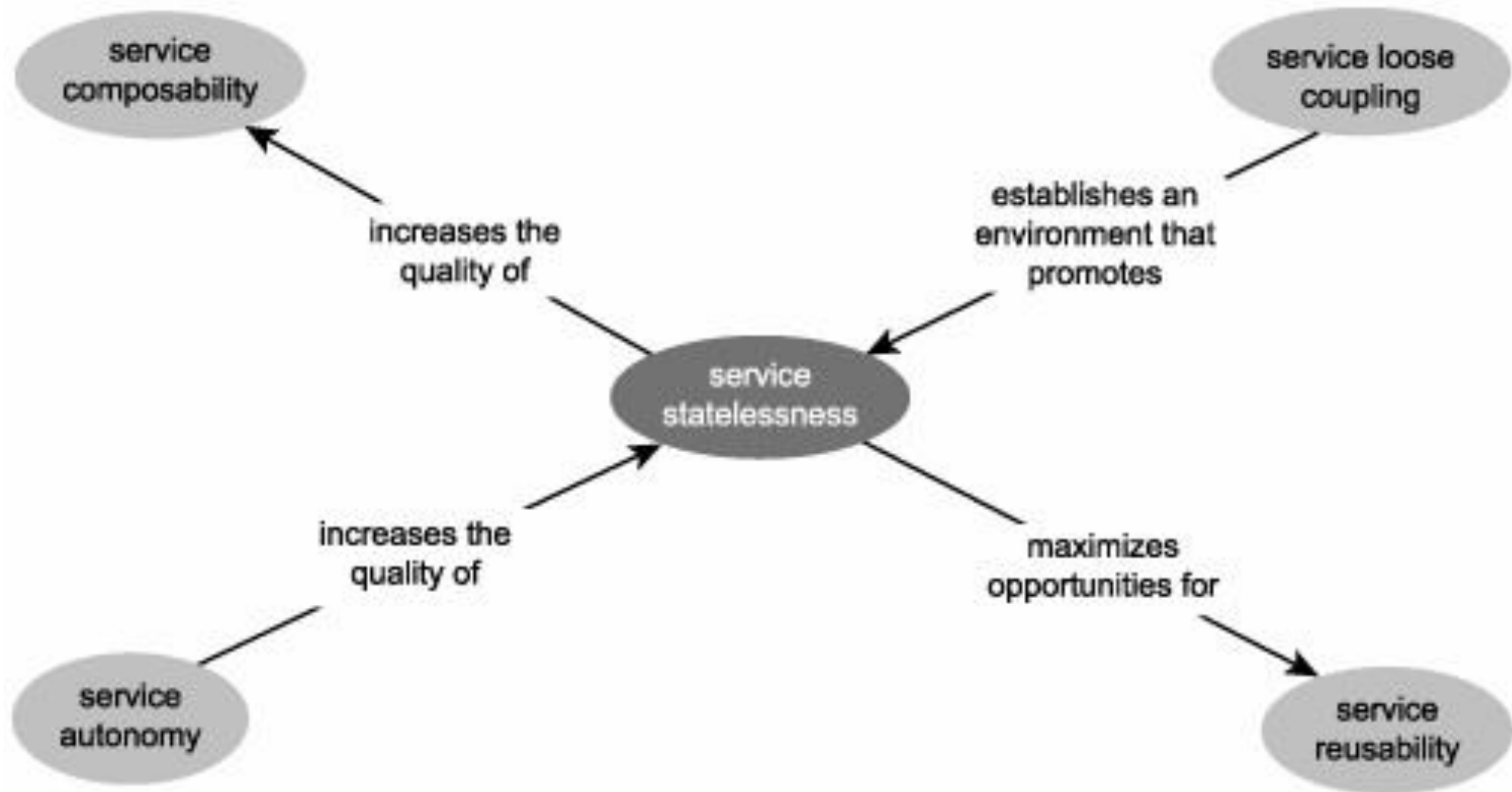
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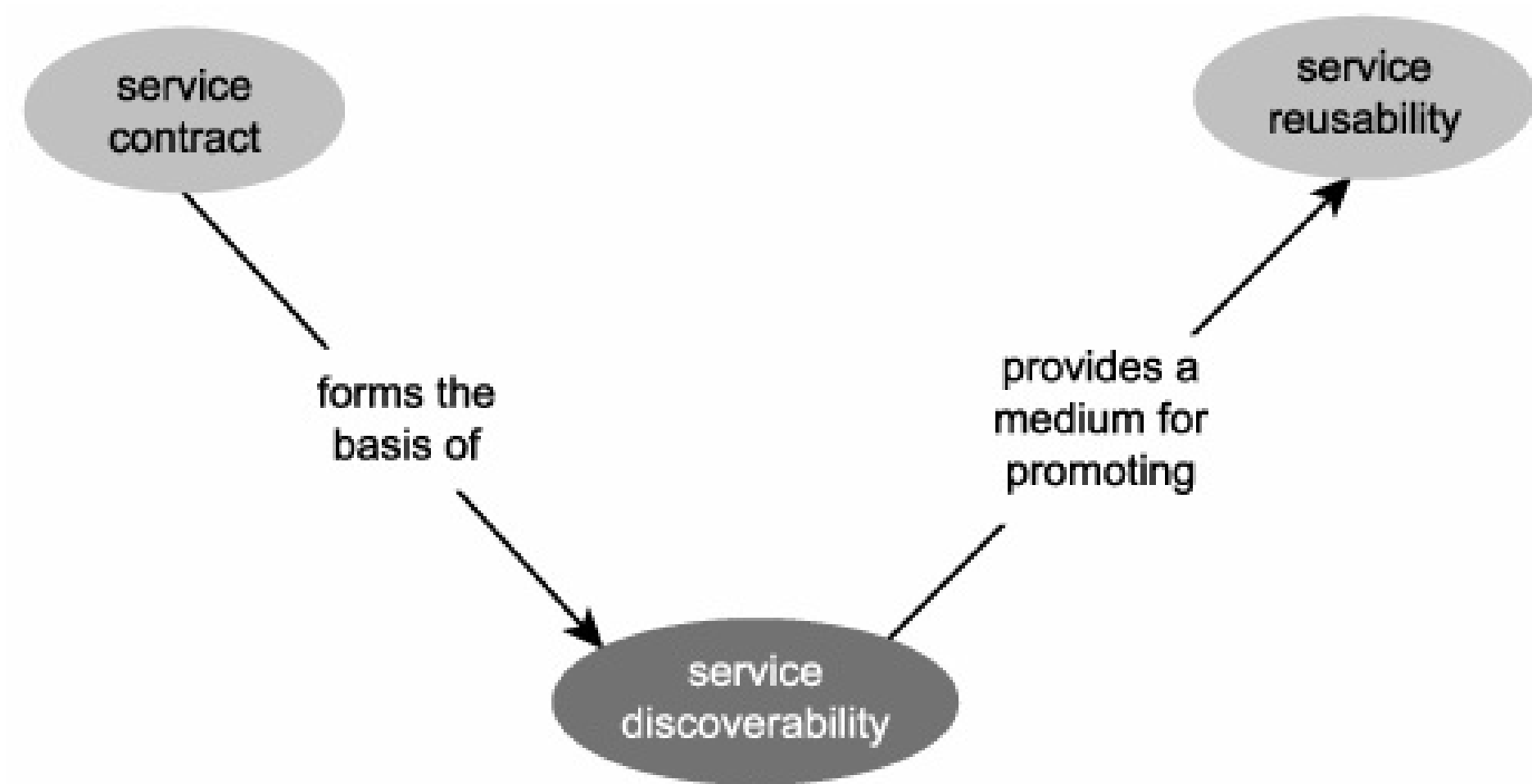
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Inter-relation between Service Orientation Principles



Service-Orientation vs. Object-Orientation

Service Orientation Principle	Related Object-Orientation Principle
Service Reusability	Reusable classes, abstraction, encapsulation
Service Contract	Interfaces
Service Loose Coupling	Interfaces
Service Abstraction	Abstraction, encapsulation
Service Composability	Association, aggregation, composition
Service Autonomy	Objects following Single responsibility principle
Service Statelessness	Objects are stateful
Service Discoverability	Interfaces

References

- “Service-Oriented Architecture: Concepts, Technology, and Design” by Thomas Erl, Pearson Education